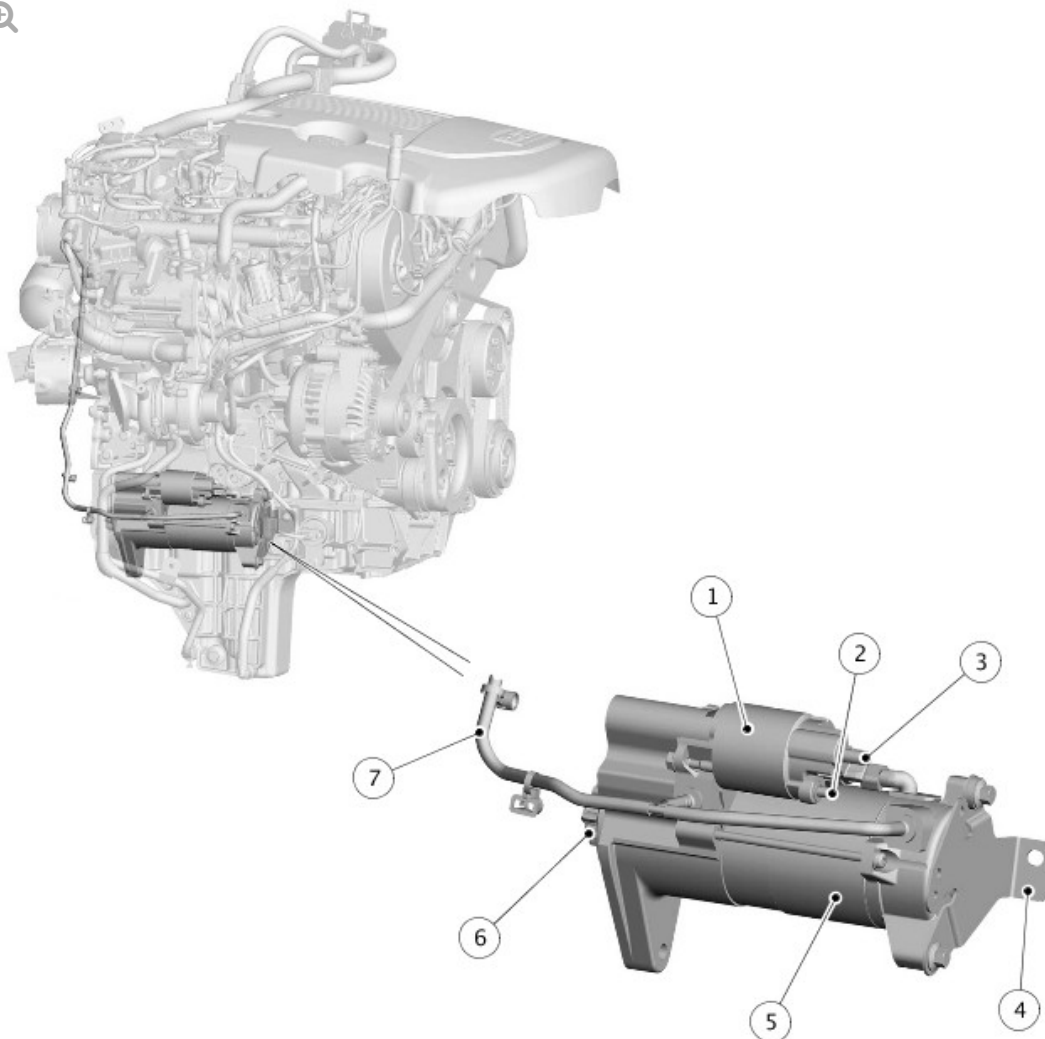


2016.0 RANGE ROVER (LG), 303-06

STARTING SYSTEM – TDV6 3.0L DIESEL

DESCRIPTION AND OPERATION

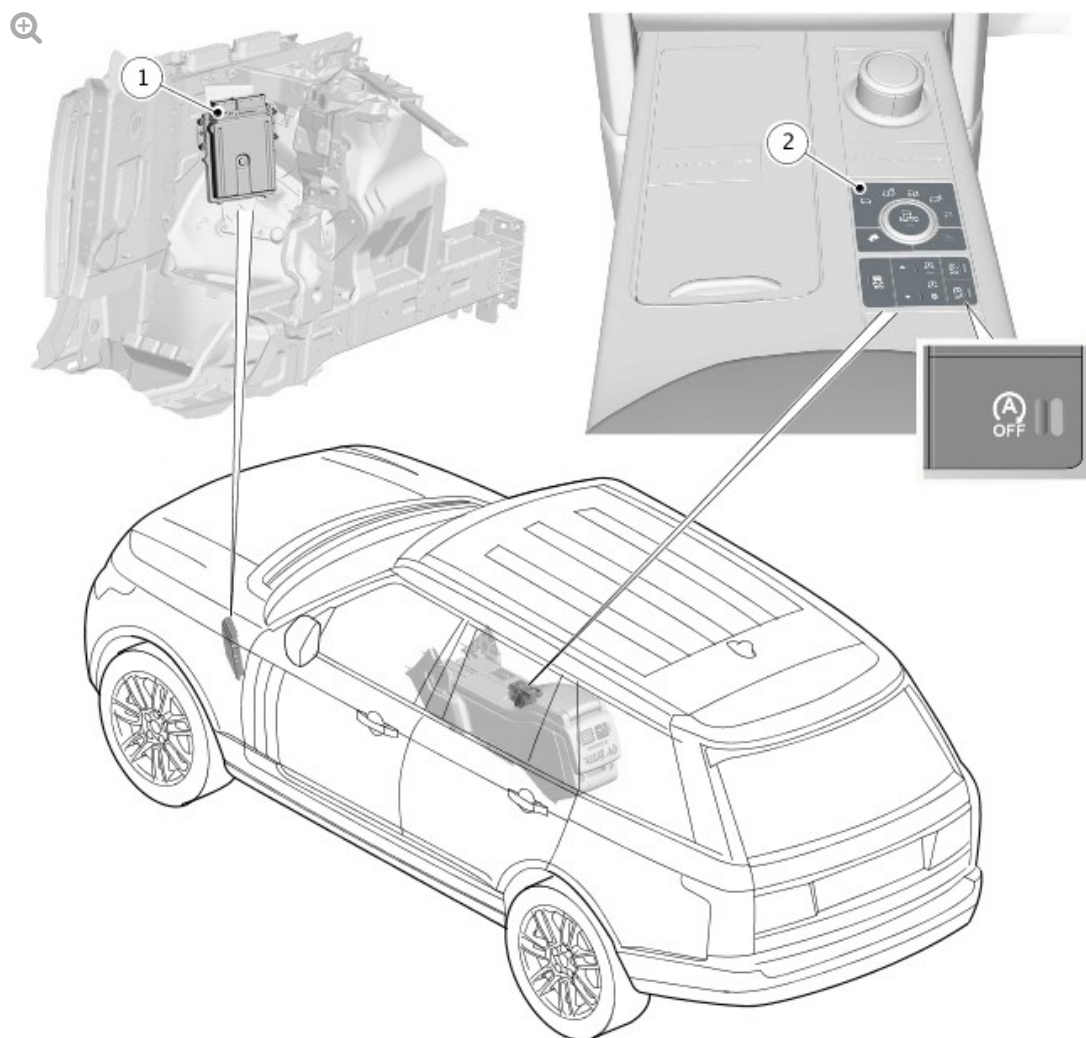
COMPONENT LOCATION - SHEET 1 OF 2 - STARTER MOTOR



E168664

ITEM	DESCRIPTION
1	Solenoid
2	Solenoid electrical connection
3	Battery electrical connection
4	Support bracket
5	Motor
6	Pinion gear
7	Breather tube

COMPONENT LOCATION - SHEET 2 OF 2
- AUTO STOP/START SYSTEM



E168839

ITEM	DESCRIPTION
1	Engine Control Module (ECM)
2	Auto stop/start switch

OVERVIEW

The starting system consists of an electric starter motor that engages with the ring gear on the crankshaft drive plate. Operation of the starter motor is controlled by the Engine Control Module (ECM).

In some markets the starting system incorporates a Auto stop/start system. The system is controlled by the ECM using Controller Area Network (CAN)

messages and signals from other system components and modules to determine the correct conditions for system operation. The Auto Stop/start system automatically stops and re-starts the engine when appropriate conditions are met. This reduces the amount of time the engine is running at idle speed which improves economy and reduces engine emissions.

The Auto stop/start system is automatically enabled each time the ignition is switched on. The driver can disable the system using the Auto stop/start switch.

The Auto stop/start system stops the engine when it is not needed. The engine will restart automatically when vehicle parameters are met and the driver releases the brake pedal.

To accommodate the increased electrical loads, vehicles fitted with the Auto stop/start system are equipped with two batteries.

For additional information, refer to: [Battery and Cables](#) (414-01 Battery, Mounting and Cables, Description and Operation).

DESCRIPTION

STARTER MOTOR

The starter motor is located on the rear right side of the oil pan. The rear of the starter motor is attached to a support bracket on the oil pan. The starter motor incorporates a snorkel breather pipe to assist with ventilation and sealing, during water submergence.

The starter motor is rated at 2.0 kW and comprises a series wound motor, an overrunning clutch and pinion, and a solenoid.

Power to operate the motor is supplied from Battery Junction Box 2 (BJB2). Power to operate the solenoid is supplied from the starter relay in the Engine Junction Box (EJB).

AUTO STOP/START SWITCH

On vehicles with Auto stop/start system, the Auto stop/start switch is integrated into the Terrain Response® switchpack in the floor console. An amber Light Emitting Diode (LED) in the Auto stop/start switch remains illuminated while the Auto stop/start system is active. Selection of the Auto stop/start switch is transmitted from the Terrain Response® switchpack to the ECM on the High Speed (HS) CAN chassis systems bus, and the HS CAN powertrain systems bus via the Gateway Module (GWM).

AUTO STOP/START WARNING INDICATOR



E168805

On vehicles with Auto stop/start system, an Auto stop/start warning indicator is located in the Instrument Cluster (IC). The warning indicator illuminates green when the engine stops during the Auto stop/start cycle, then goes off when the engine restarts. The warning indicator illuminates white when the Auto stop/start system is disabled. The warning indicator is controlled by a HS CAN powertrain systems bus message from the ECM.

OPERATION

VEHICLES WITHOUT AUTO STOP/START SYSTEM

When the Engine stop/start switch is operated, if the Keyless Vehicle Module (KVM) detects a valid smart key in the vehicle the Central Junction Box (CJB) transmits a hardwired crank request to the ECM.

For additional information, refer to: [Anti-Theft - Passive](#) (419-01B Anti-Theft - Passive, Description and Operation).

When the ECM receives a crank request from the CJB it energizes the starter relay, provided the Transmission Control Switch (TCS) is in PARK ('P') or NEUTRAL ('N') and the brake pedal is pressed. The energized starter relay supplies battery power to the starter solenoid, which energizes and causes the pinion gear to engage with the ring gear. When the starter solenoid is energized it also closes high-current contacts, which connects battery power from BJB2 to the motor to turn the engine.

VEHICLES WITH AUTO STOP/START SYSTEM

During a drive cycle, with the Auto stop/start system enabled and the TCS in DRIVE ('D') or SPORT ('S'), the ECM performs an auto stop when one of the following occurs:

- The vehicle stops from a speed greater than 4 km/h (2.5 mph) and sufficient brake pressure is applied to the brake pedal to ensure the vehicle is stationary
- The vehicle is stationary and 'N' is selected on the TCS.

NOTE:

On a gradient an auto stop will only occur if the driver applies a minimum of 7 bar (101.5 lbf/in²) + / - 3 bar (43.5 lbf/in²) brake pressure and will not occur at all on gradients about in excess of 1 in 4.34 (23%).

The green Auto stop/start warning indicator in the IC illuminates to advise that the engine has shut down. Other warning indicators normally associated with an engine shut down, for example the ignition and low oil pressure warning indicators, are suppressed so will not illuminate during an auto stop.

During an auto stop, the ECM operates the fuel pressure control valve to

maintain a fuel rail pressure of 250 bar (3625 lbf/in²), instead of releasing the fuel pressure as normally occurs at engine shut down. This ensures there is adequate fuel volume and pressure for an instant auto restart.

The following conditions will prevent an auto stop:

- A steering wheel paddle switch has been used to select a gear (transmission Commandshift mode)
- The ambient air temperature is outside the range of 2 to 40 °C (36 to 104 °F)
- The ambient air pressure is below 720 hPa
- The engine coolant temperature is outside the range of 20 to 107 °C (68 to 224 °F)
- Brake vacuum is less than the threshold value
- The hood is open
- The driver door is open
- The driver seatbelt is not fastened
- High demand from the climate control system requires the engine to be running, for example windshield demist selected
- The battery charge is low
- A Terrain Response® special program is active
- Hill descent control is active
- The level in the fuel tank is too low (approximately 6 liters remaining in tank)
- Active regeneration of the Diesel Particulate Filter (DPF) is required or in progress
- Auto stop/start system is disabled using the Auto stop/start switch.

While the vehicle is stationary, if a condition that prevented an auto stop is removed the ECM automatically performs an auto stop.

The ECM automatically restarts the engine when one of the following occurs:

- The brake pedal is released with 'D' or 'S' selected on the TCS.
- The accelerator pedal is pressed
- A paddle switch is used to select a gear
- REVERSE ('R') is selected on the TCS
- Climate control system demand increases
- The vehicle speed exceeds approximately 1 km/h (0.5 mph)
- The battery charge becomes low
- Brake vacuum reduces below the threshold value
- Auto stop/start system is disabled using the Auto stop/start switch.

NOTE:

When the brake pedal is released during an auto restart the ABS (Anti-lock Brake System) control module briefly applies the brakes to hold the vehicle stationary until the engine has restarted.

The following will prevent an auto restart, and require use of the engine stop/start switch to restart the engine:

- The hood is open
- The driver seatbelt is not fastened.

AUTO STOP/START STRATEGY

Engine Speed up to 330 RPM:

When the engine speed is less than 330 RPM, and an auto stop is initiated, the restarting of the engine is controlled via the starter motor. The starter motor has the capability of engaging with the transmission ring gear whilst still rotating. When the starter motor rotates the engine speed up to 330

RPM, the ECM then activates the fueling and engine systems to restart the engine.

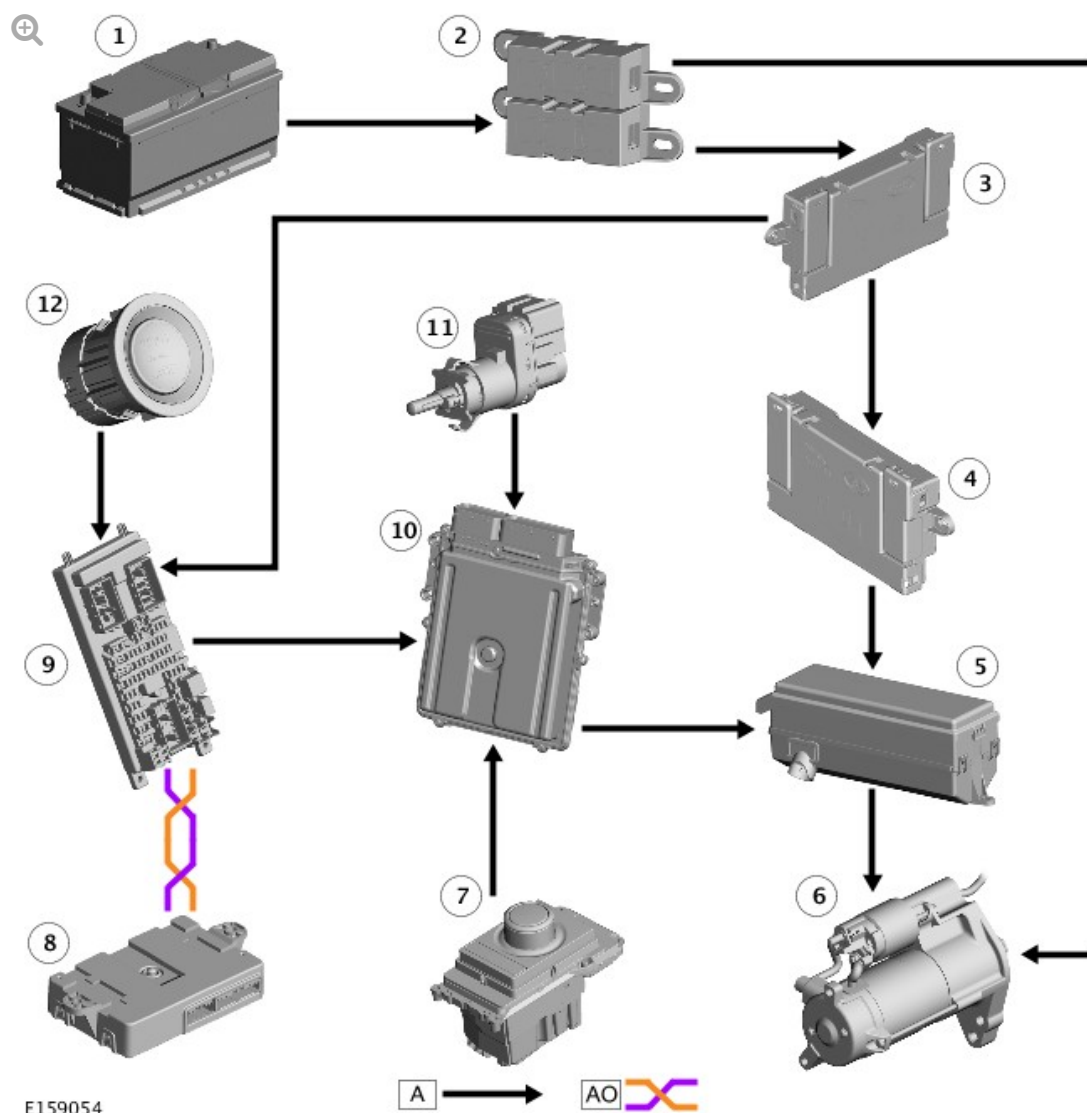
Engine Speed above 330 RPM:

When the engine speed is above 330 RPM, and an auto stop is initiated, the restarting of the engine is controlled by the ECM. The ECM activates the fueling and engine systems only.

CONTROL DIAGRAM - SHEET 1 OF 2 - VEHICLES WITHOUT AUTO STOP/START

NOTE:

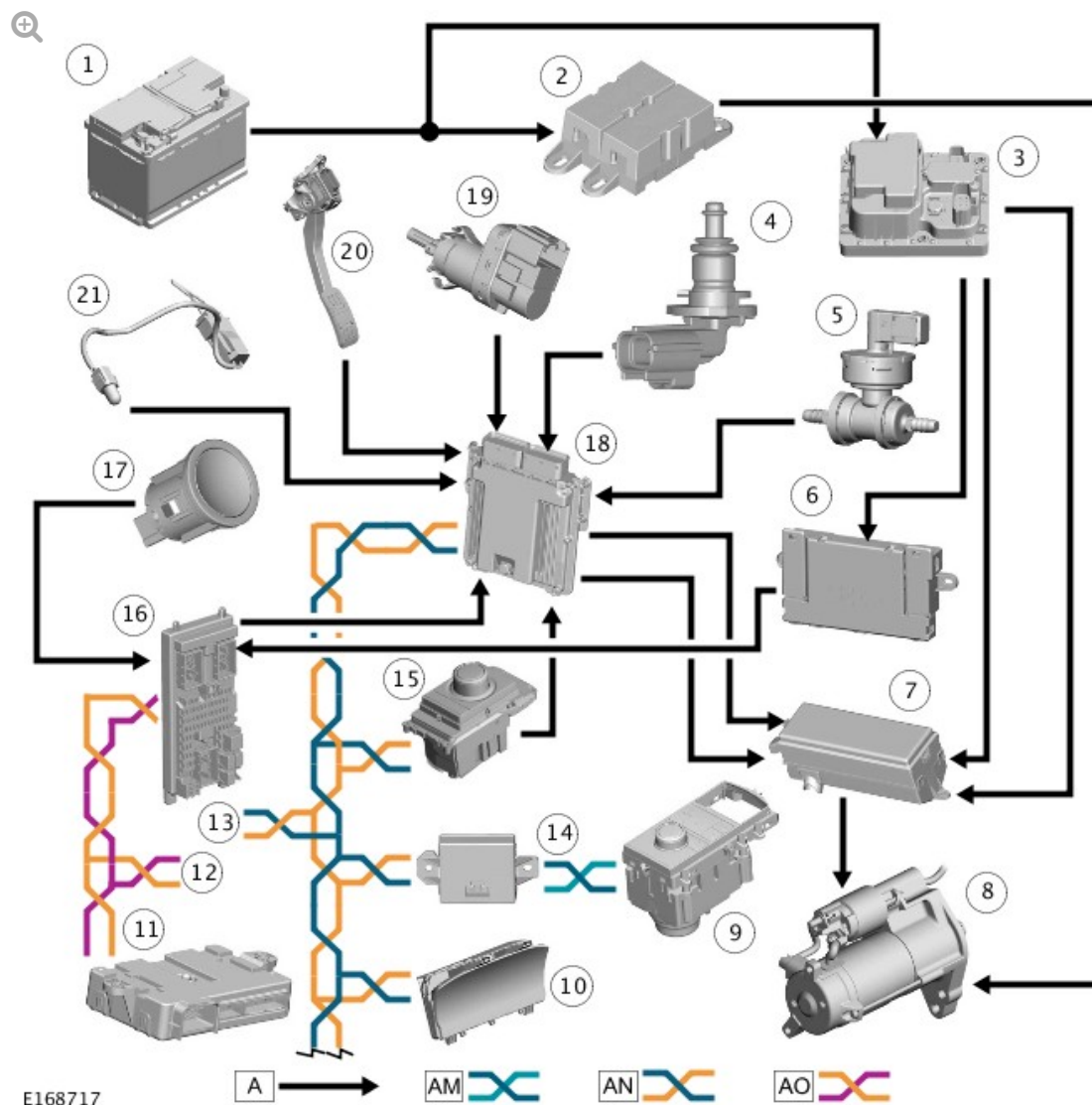
A = Hardwired; **AO** = Medium Speed (MS) Controller Area Network (CAN) body systems bus; **AM** = High Speed (HS) CAN chassis systems bus; **AN** = HS CAN powertrain systems bus.



ITEM	DESCRIPTION
1	Battery
2	Battery Junction Box 2 (BJB2)
3	Battery Junction Box (BJB)
4	Auxiliary Junction Box (AJB)
5	Engine Junction Box (EJB)
6	Starter motor
7	Transmission Control Switch (TCS)
8	Keyless Vehicle Module (KVM)
9	Central Junction Box (CJB)
10	Engine Control Module (ECM)

11	Brake pedal switch
12	Engine stop/start switch

CONTROL DIAGRAM - SHEET 2 OF 2 - VEHICLES WITH AUTO STOP/START SYSTEM



ITEM	DESCRIPTION
1	Battery
2	Battery Junction Box (BJB2)
3	Power Supply Distribution Box (PSDB)
4	Engine Coolant Temperature (ECT) sensor
5	Brake vacuum sensor
6	Battery Junction Box (BJB)

7	Engine Junction Box (EJB)
8	Starter motor
9	Terrain Response® switchpack
10	Instrument Cluster (IC)
11	Keyless Vehicle Module (KVM)
12	Connection to other control modules on Medium Speed (MS) Controller Area Network (CAN) body systems bus
13	Connection to other control modules on High Speed (HS) Controller Area Network (CAN) powertrain systems bus
14	Gateway Module (GWM)
15	Transmission Control Switch (TCS)
16	Central Junction Box (CJB)
17	Engine stop/start switch
18	Engine Control Module (ECM)
19	Ambient Air Temperature (AAT) sensor
20	Accelerator pedal
21	Brake pedal switch